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Second-generation Drones

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3.1 Introduction

The second generation of drones represents a pivotal era in the evolution of unmanned aerial vehicles (UAVs), marking the transition from basic, manually operated machines of the early 20th century to more advanced, radio-controlled and semi-autonomous systems. Spanning from the 1950s through the early 2000s, this generation was characterized by rapid advancements in electronics, navigation, propulsion, and communication technologies, transforming drones from simple military training tools into versatile aerial platforms with applications in defense, science, and industry (Kardasz, et al. 2016).

During this period, drones evolved from analog radio-operated prototypes to aircraft equipped with digital control systems, real-time telemetry, and global positioning system (GPS) technology, enabling precise navigation and mission repeatability. The introduction of flight stabilization systems, onboard sensors, and data transmission links allowed drones to operate more efficiently and with greater accuracy, significantly expanding their operational capabilities. These innovations made UAVs invaluable not only in military reconnaissance and surveillance but also in civilian domains such as environmental monitoring, aerial mapping, and scientific research.